

Sensors, Microcontrollers & Power Components

Microcontrollers/Boards:

- Cellular + Computation Boards:

- Computation Board: [Firebeetle ESP32 IOT Board](#) - \$12.9
 - Sensor Breakout Stack: [DF Robot Gravity Sensor Expansion Board](#) - \$4.9
 - Cellular + GPS Module: [Waveshare Cat-1/GNSS HAT for Raspberry Pi, Based On SIM7670G module \(part no: 26861\)](#) - \$44.49
 - GNSS Antenna: [GNSS Multiband antenna ANN-MB-00](#) - \$60.00
 - Cellular Antenna: [Proxicast Ultra Low Profile MIMO 4G / 5G](#) - \$64.95
 - SMA-->u.FL: [Waterproof SMA to u.FL RF Adapter](#) - \$5.90
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Sensors:

- Ph Sensor (Analog): [DF Robot Industrial Grade Ph](#) - \$64.9
- DO Sensor (Analog): [DF Robot Dissolved Oxygen Sensor](#) - \$169.0
- EC Sensor (Analog): [DF Robot Electrical Conductivity Sensor](#) - \$199 or order from [Digikey](#)
- Turbidity Sensor (Analog): [DF Robot Turbidity Sensor](#) - \$9.9
- Temp Sensor (Digital): [DF Robot Temp Sensor](#) - \$7.50
- ORP Sensor (Analog): [DF Robot Oxygen Reduction Potential Sensor](#) - \$129
- Water Leak (Digital): - [UHPPOTE Wired Water Liquid Leakage Probe Detector Leak for Alarm System](#) - \$15.30

- Derived Measurements:

1. **TDS (mg/L)** = $0.64 * EC (\mu S/cm)$
2. **Algae Bloom Risk Score** = weighted sum of (high temp + high turbidity + pH↑ + DO fluctuations).
3. **Nutrient Enrichment Indicator** = EC↑ + ORP↓ + DO↓ trends after rainfall/runoff events.
4. **Organic Pollution:** Low DO, low ORP, and high turbidity indicate decomposing organic matter; pH and temperature affect bacterial activity; DO decline over time estimates BOD.
5. **Thermal Pollution:** Rising temperature with falling DO signals heated discharges or runoff.
6. **Chemical/Industrial Pollution:** Extreme pH, high EC, abnormal ORP, and suppressed DO reveal ionic or chemical effluent, though specific compounds remain unknown.

7. **Eutrophication/Algal Blooms:** Elevated daytime DO, nighttime DO drops, higher pH, and turbidity growth show algal photosynthesis and bloom intensity.
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Power Components:

- Solar Panels:

1. (x2): [10.6 x 13.4 10W 12V Solar Panel](#) - \$38.75

- Lipo Battery: [Li-ion 3.7 V nominal, ≥6000 mAh](#) - \$24.99

- Solar charger: [DF Robot Solar Manager](#) - \$14.95

- ESP32 Voltage Reg: [Pololu D24V10F3](#) - \$12.95

- Ada FONA Voltage Reg: [AITRIP DC Adjustable Buck](#) - \$14.99

- Ada FONA Input Capacitor: [Radial Leaded 25VDC 470uF Capacitor](#) - \$0.78